NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

TECHNICAL SPECIFICATION FOR

Vertical Lift Fabric Door

VERTICAL LIFTING FABRIC DOOR SYSTEM:

A. **General**: Vertical Lifting Fabric Doors door is designed in such a way that the material, construction, and operation of the door shall conform to the performance specifications contained herein and it doesn't consist of any protruding parts affecting personal safety in site. Door operate properly without binding, interference, or damage to the adjacent structure

B. **Door Leaf:** The door leaf is made of two layers of very strongly vinyl coated polyester fabric weighing approximately 700g/sq.m), capable of carrying 2400N/5cm. It shall be UV stabilised, self extinguishing (0 - 25 flame spread), and suitable to withstand temperatures between – -30°C to +70°C. The translucent material if used, shall also meet the same requirements and loading as the standard door fabric. The fabric is attached to both sides of the intermediate beams, top beam and bottom beam with self tapping screws through corrosion resistant aluminium batten strips Powder coated clamp strips to match the color of the fabric. The door fabric shall be pulled tight between the intermediate beams when the door is fully closed and wind lock engaged. The fabric have been tested to meet the criteria of attached test documents and standards as below:

- **Flame retardency:** tested according: EN 13501-1, B-s2,d0.
- **Tensile strength:** 2500/2000 N / 5 cm acc to EN ISO 1421-1
- **Tear resistance:** 400 / 300 N acc to DIN 53363
- **Light resistance:** ≥ 7 acc to ISO 105-B02
- **Radar reflecting coefficient:** 0,3dB, ~0,1%

C. **Door Color:** manufacturer's standard color, Red, Blue, Grey, White, White aluminum, Tan, Green, Graphite or Translucent.

D. **Doors:** consist of hoist up fabric doors with intermediate aluminium beams or trusses. Fabric gathered above the head of the opening. Regardless of location, the minimum design wind load will be to 1.1 kPa.

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E. **Door Beams:** The intermediate beams are corrosion resistant extruded aluminium tempered after the extrusion process to ensure required strength and have a length to depth ratio not to exceed 100:1 to meet the required wind load, calculated according EN 13241-1 with beam safety factor 1.5 X EN 12604. They shall be spaced 700 to 1400 metres apart, dependent upon the wind load. At each end of the beams there shall be a guide block of self-lubricating nylon material, or rollers that run along the guides. The intermediate beams must have a minimum depth of 290mm. The intermediate door beams and guide block or rollers shall be designed to carry the full design wind load without failure or being pulled from the door guides. There shall be no multiple lifting cables and holes distributed along the length of the intermediate beam.

F. **Bottom Section:** The bottom section is made of steel. A rubber seal to be fixed to the section to ensure tightness against the floor /ground. Bottom section prevent lifting of door during wind load and secured at locked position with an automatic wind lock system.

G. **Door Guide:** The vertical guides is an integral part of the door, made of extruded aluminium with a suitable depth and width dependent on the size of the intermediate beams. The guides are designed to provide weather sealing on the inside and the outside faces. The guides are designed to provide sufficient strength to transfer the wind load and safety arrestor forces from the door beams, at maximum wind load, to the jamb steel support system. The guides are easily replaceable in case of damage. The installer shall furnish jamb structural steel fasteners of size and type required for attachment of the guide rails as shown on the approved shop drawings.

H. **Drive Units:** The door leaf have an IEE / EN labelled single or dual motor drive system with torque sized as appropriate for the weight of the door leaf. The gear motor equipped with a drum on which the belts/ropes are wound. If a single motor drive is used, the belts/ropes shall be wound on the same drum. The belts/ropes will be attached to the bottom beam via the safety arrestors.

I. **Bottom Beam:** The bottom beam designed with a suitable width and depth to carry the load of the intermediate beams when the door is open, and to ensured full closing and a tight floor seal in heavy winds. A heavy U-shaped bottom rubber seal is provided to form a tight seal with the floor, even on uneven surfaces.

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J. **Self-Activating Wind Lock:** Each door leaf will be provided with two self activating wind locks which activate and lock the bottom beam when the door reaches its closed position. This locking action shall act to maintain a tight floor seal and intermediate stability even under conditions of heavy wind. A proximity switch, with indicator shall be provided to notify the door operator that all wind locks are engaged properly.

K. **Control Panel:** The control panel is IP 65 rated.

L. **Wind Resistance:** Doors are engineered to take into account the local environmental conditions i.e. high winds, high temperatures, high levels of dust and high humidity levels.

M. **Water penetration:** Doors are engineered in accordance to EN12425 Class 3 with resistance to water penetration of 110PA for a closed door with or without pass door.

N. **Air Tightness:** Doors shall be engineered in accordance to EN 12426 **Class 2- 3,** with or without pass-door.

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